How Small Can a Neural Network Be?

Context

- MNIST is a standard image classification benchmark
- Many models achieve >99%
- Most models are heavily overparameterized

Research Question

What is the minimum-capacity neural network that achieves ≥97% test accuracy on MNIST?

Definition

Model size = number of trainable parameters

Competitive accuracy = ≥97% test accuracy

Why Does This Matter?

Efficiency vs. performance trade-off

- Smaller models:
  - Faster training
  - Lower memory usage
  - Less energy consumption
  - Better for edge devices
  - Helps understand model capacity and generalization

Hypothesis

We expect MNIST can reach 97% accuracy with **significantly fewer parameters** than commonly used architectures.

Experimental Plan

1. Architecture Search

- Vary width (hidden units)
- Vary depth (number of layers)
- Compare parameter counts

2. Evaluation

- Fixed training setup
- Same optimizer & hyperparameters
- Measure test accuracy
- Repeat runs to avoid randomness effects

3. Output

- Plot: Parameters vs Accuracy
- Identify smallest model ≥97%